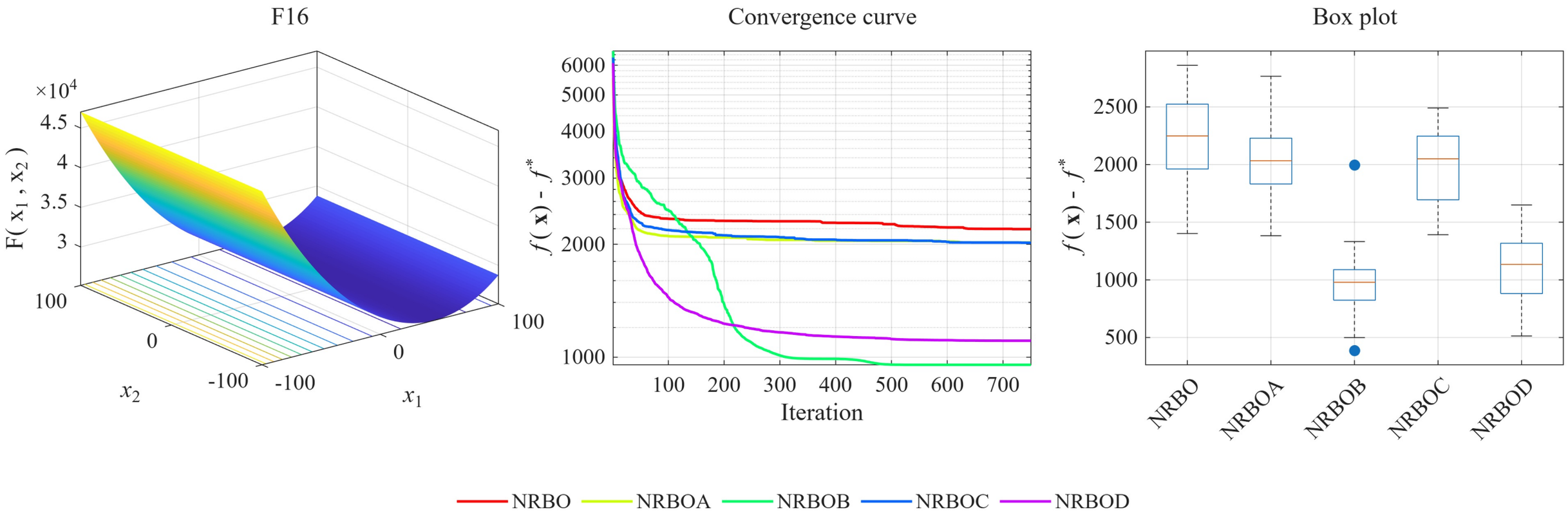

F16
×10^4
4.5
4
3.5
3
F(x1 , x2)
100
0
-100
x2
-100
0
x1
Convergence curve
6000
5000
4000
3000
2000
1000
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
2500
2000
1500
1000
500
f(x) - f*
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD